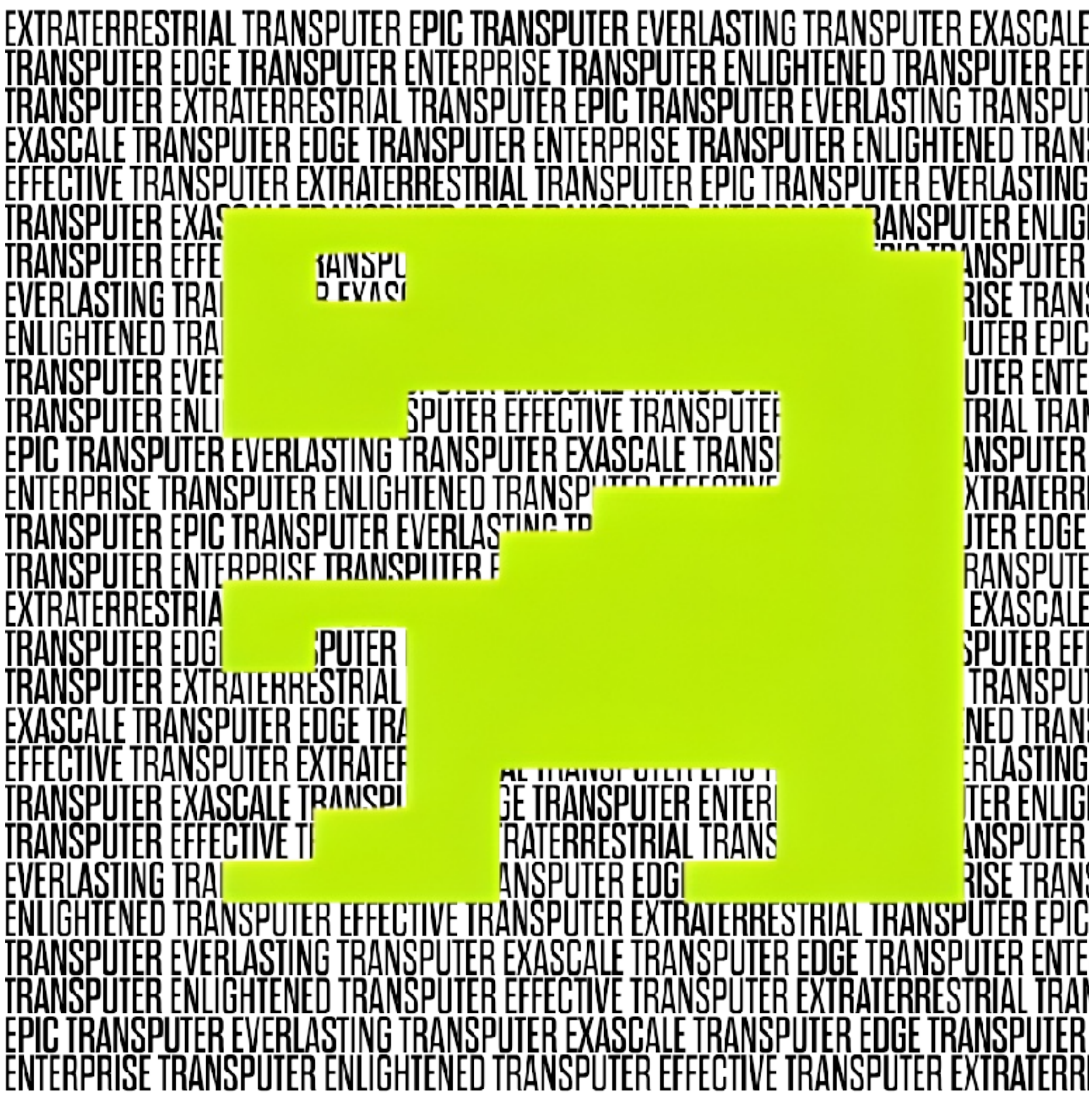


Introducing ET

The fully open source
manycore platform.

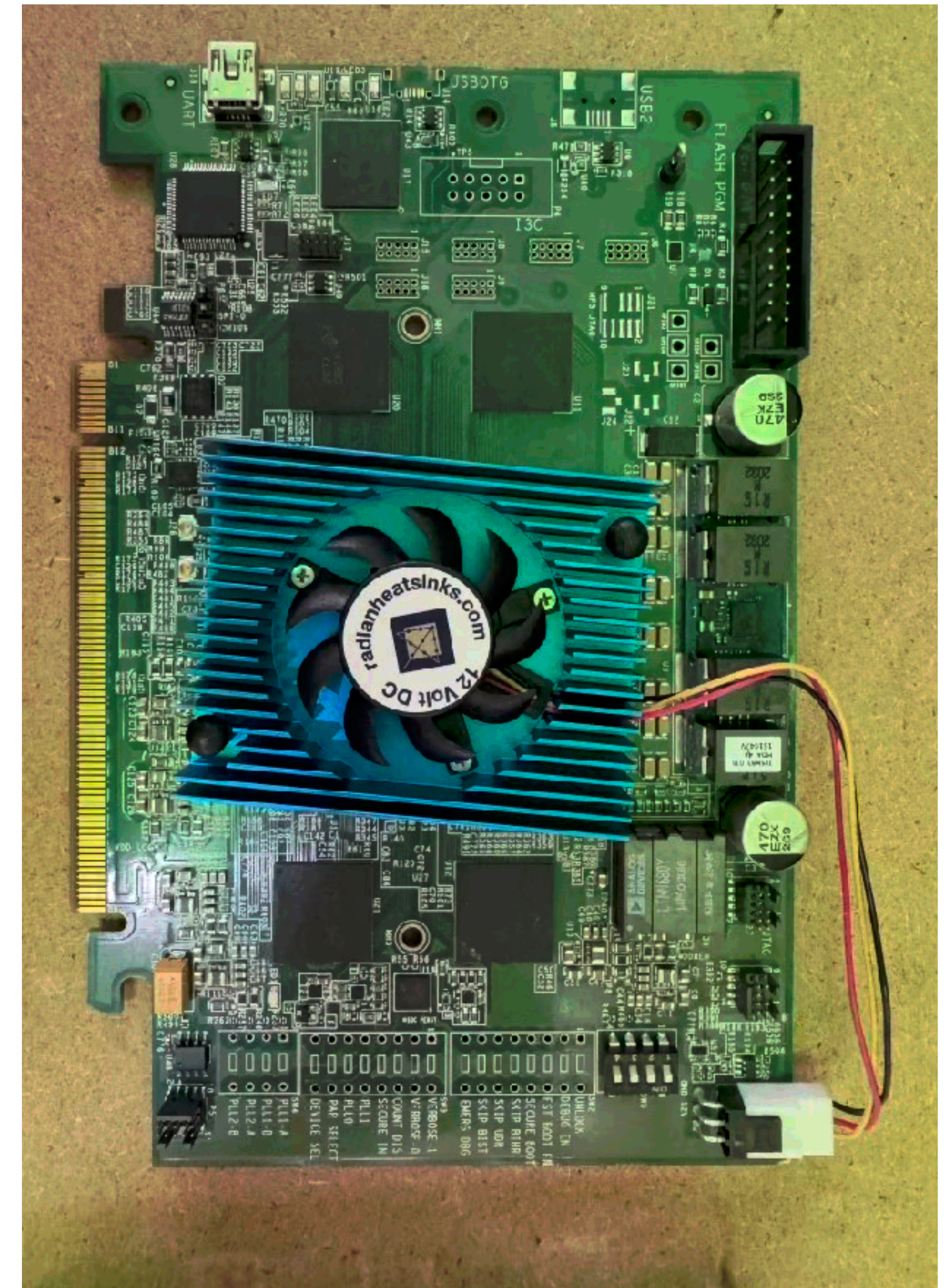
Gianluca Guida, 10/28/2025



What is ET?

The fully open source ~~AI accelerator~~ *manycore platform*.

- Based on Esperanto's ETSOC-1 chip.
- 1088+ RISC-V cores with custom vector and tensor acceleration at 300-800MHz.
- 40W consumption, no external power needed.
- **Ainekk** is *open-sourcing it* under the **AI Foundry** umbrella.



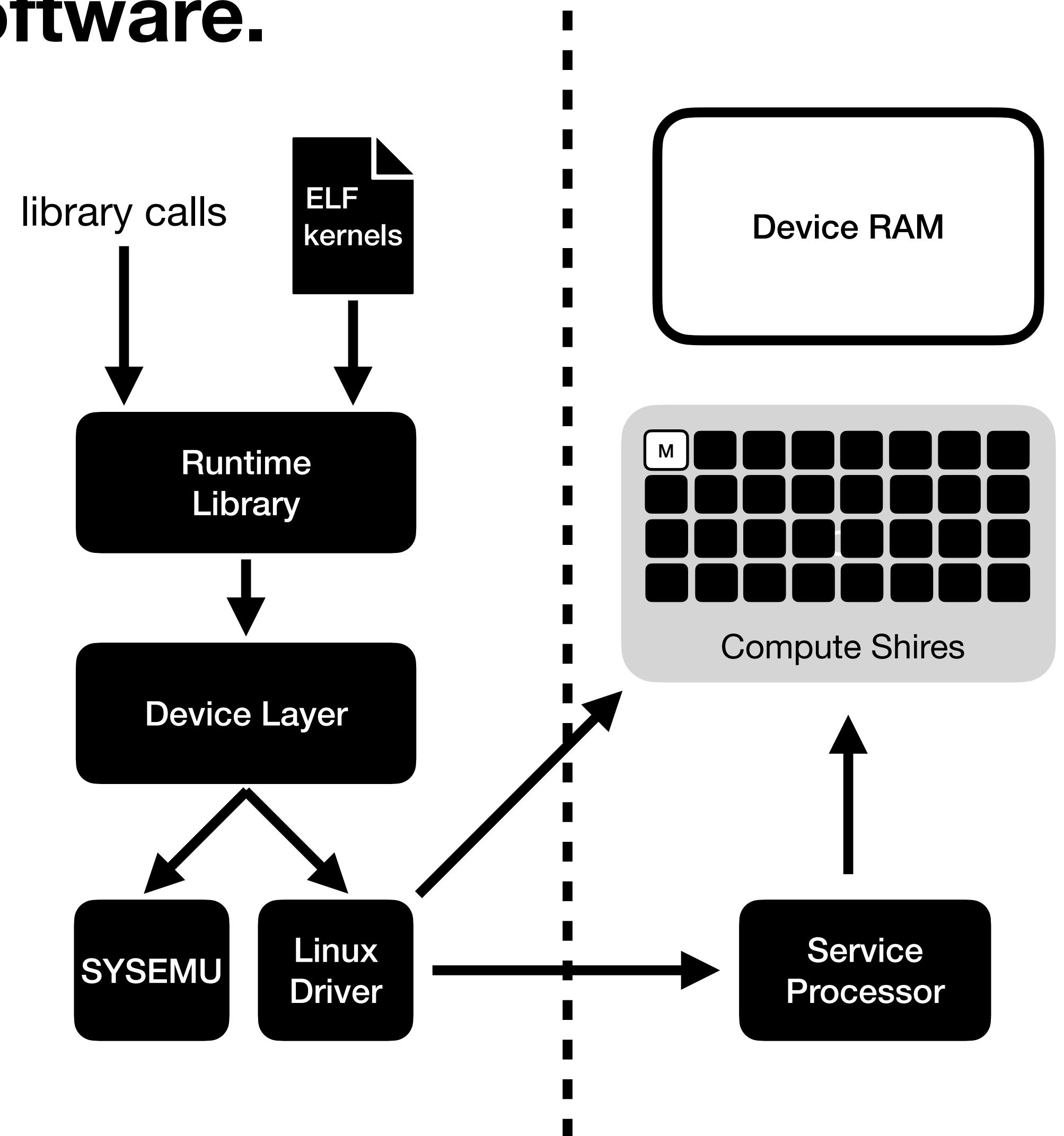
AI Foundry GitHub repos

- The ETSOC1 Programmer's Reference Manual
 - <https://github.com/aifoundry-org/et-man>
- GCC Port for Minions
 - <https://github.com/aifoundry-org/riscv-gnu-toolchain>
- The ET Platform software
 - Runtime to firmware (emulation or real hardware)
 - <https://github.com/aifoundry-org/et-platform>
- llama.cpp port to ETSOC1
 - <https://github.com/aifoundry-org/llama.cpp>

ET Platform Software

A hand-wavy overview of current software.

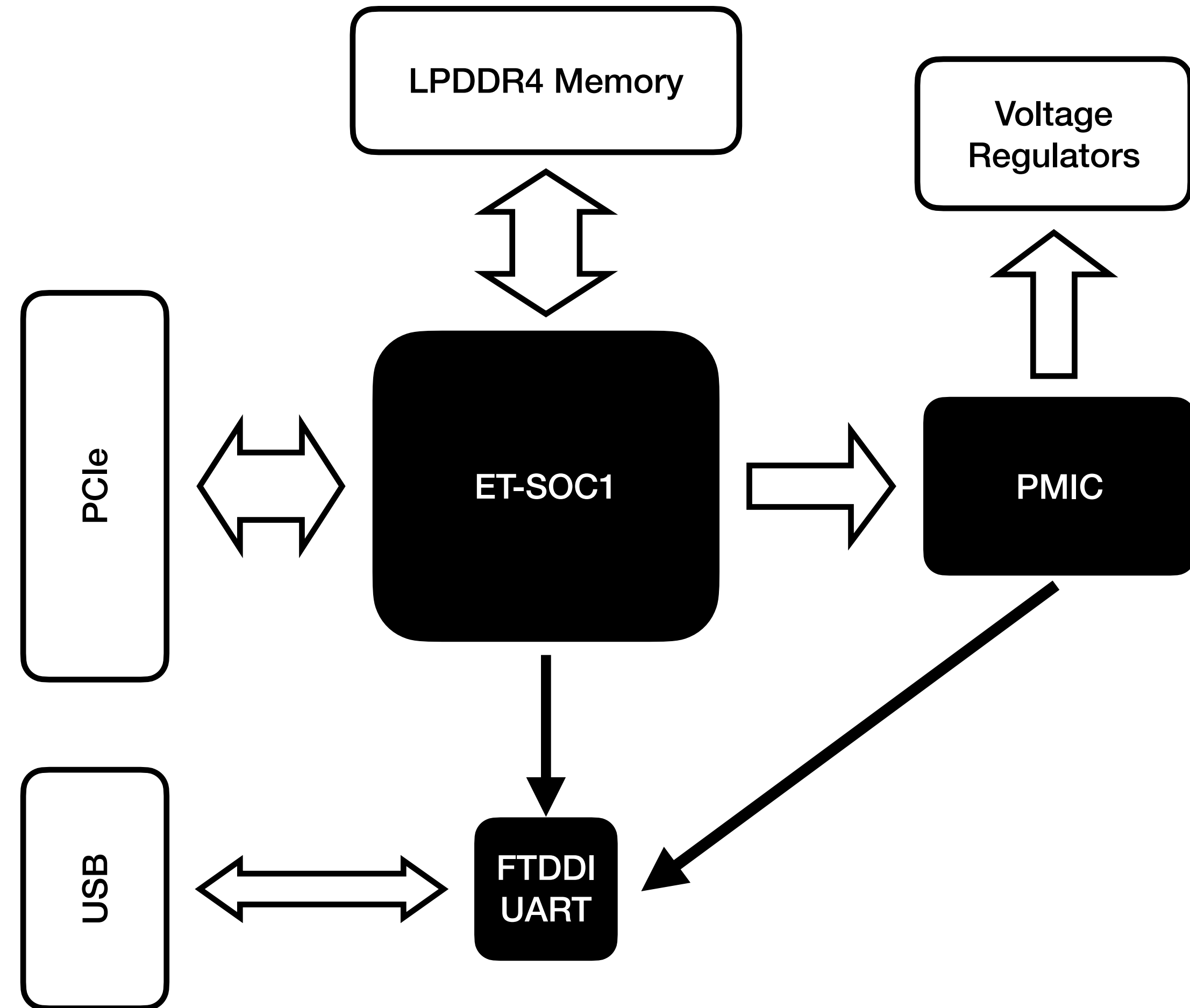
- The highest level entry point is the **runtime library**
 - Has concept of streams, events, and load and execute ELF on compute cores.
- **device layer library** abstracts access to physical devices, allowing it to emulate.
- The kernel **driver** exposes two interfaces:
 - `etX_mgmt`: for commands to control and manage the device.
 - `etX_ops`: for executing kernels on shires.
- Low level interface is message based, and requires listening, rather than call and return.



ETSOC1 Hardware

The PCIe board: Overview

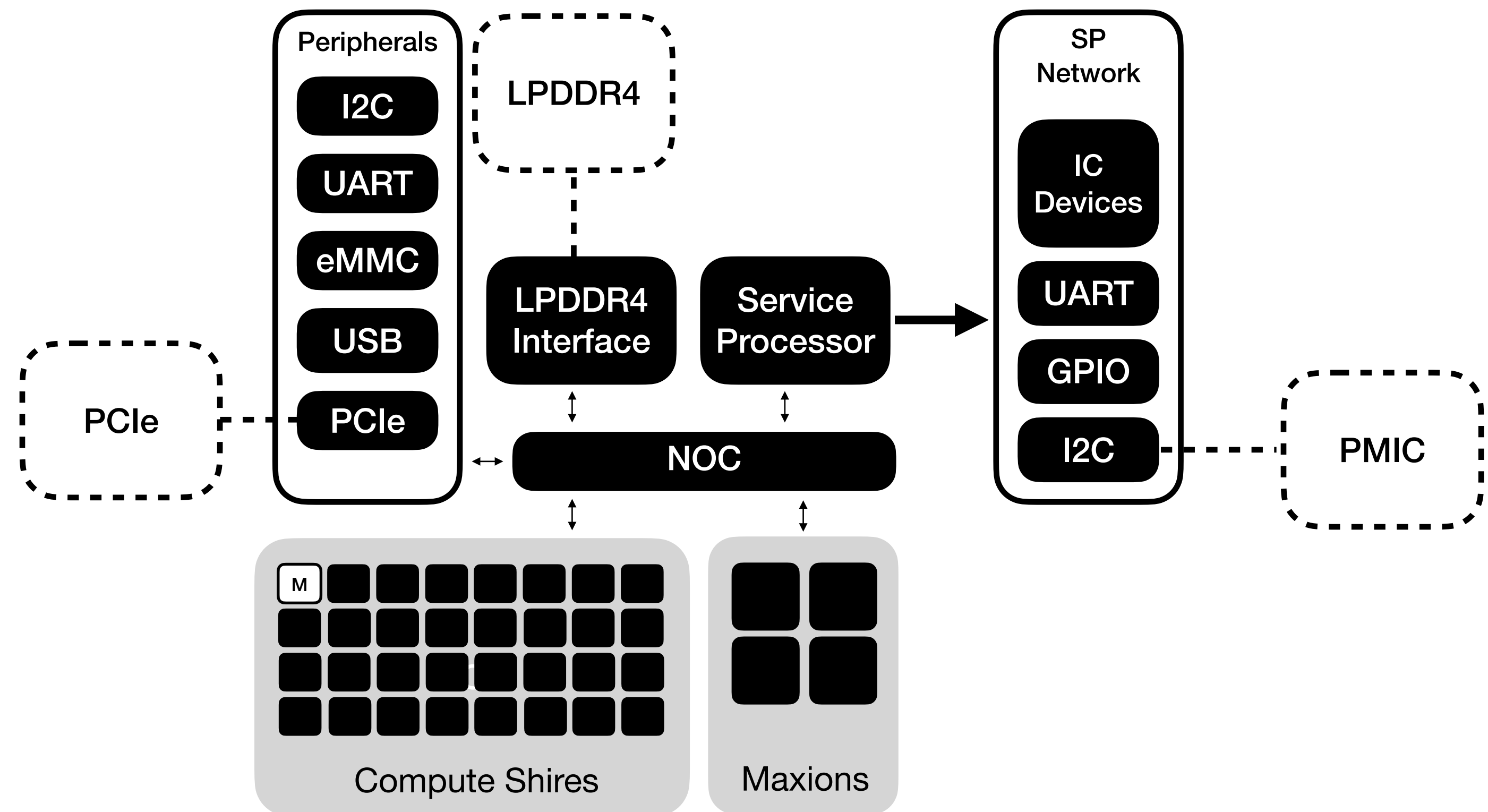
- Three main visible components:
 - ET-SOC1
the Esperanto IC
 - PMIC
Power management microcontroller
 - On board RAM (LPDDR4)
- Externally, the board has a few USB devices:
 - FTDDI UART (PMIC, SP and Master Minion serial I/O)
 - ETSOC-1 USB OTG: can be both device or host, software controlled
 - ETSOC-1 USB Device (Debug Port + Generic USB Device)



ETSOC1 Hardware

ETSOC-1: Overview

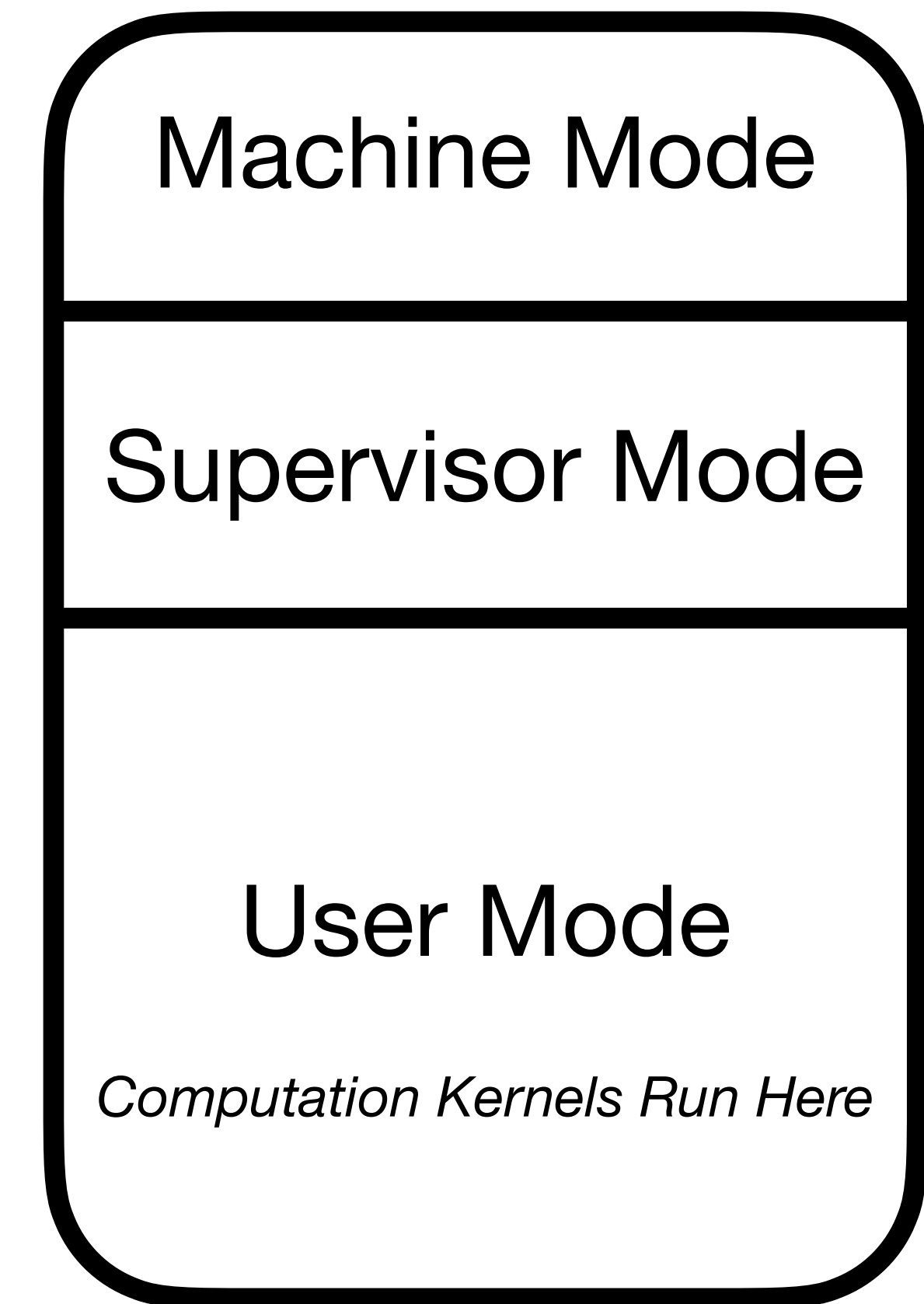
- The Service Processor controls everything.
 - Receive messages from host and execute operations.
 - Has private devices it can access (SP Network).
- Compute shires are the stars of the show:
 - 34 compute shires.
 - 32 cores (*minions*) per shire.
 - 2 HARTs (hardware threads) per *minion*.
- One shire is the Master Shire
 - Receives and sends messages.
 - Schedule work to the rest of compute shires.
- *Maxions* are out-of order RISCV64 cores
 - On the same NoC of the minions.
 - Bigger and more powerful than *minions*.



ETSOC1 Hardware

Minion: an overview

- RISC-V64 CPU, in-order.
 - `rv64imfc`
- Esperanto Extensions
 - atomics
 - packed float and integers instructions
 - tensor operations
- Contains all the RISC-V execution modes: M, S, U (no virtualization)
- Service Processor is a minion without vector/tensor extensions



How can I take part in this project?

This is fun.

- This is really *open-source hardware acceleration*.
 - ETSOC1 Manual online, all software and firmware online. And more is coming.
 - You can run (*slowly*) on the simulator even if you don't have hardware.
- The architecture is *almost completely* software defined!
 - You have 1088 RISC-V cores on your PCIe bus. What can you do with it?
- All development happening in the open, on GitHub.
 - Existing known bugs will be posted soon. Discussions and pull requests very welcome.